

10 MAINTENANCE

10.1 Maintenance Preventive

This section of the manual is dedicated to instructing and assisting operators and maintainers on procedures that ensure full machine operation, extend machine life, minimize downtime and improve overall performance. Preventive maintenance procedures must be strictly followed and treated as a priority.

All Raumak machines are designed so that they can be operated and maintained by the user. It is recommended that a preventive maintenance program be implemented, involving the following procedures.

10.1.1 Procedure for maintenance

The following maintenance tasks must be performed periodically:

- Clean the equipment daily;
- Check daily the wear and tear of the welding components such as teflons and resistors;
- Verify daily the wear and tear of the teflons on the forming tube;
- Lubricate the bearings monthly, using grease nipples;
- Check the tensioning of the belts in general (if any) daily;
- Replace the filtering element in the pressure regulator filter when necessary;
- Check daily the fastening of moving parts, if necessary retighten screws or parts;
- Check and purge filters and separators that contain liquids drained from the pneumatic system daily or as needed, as this factor will depend on the environment where the equipment is being operated.



Before starting any maintenance procedure, the person responsible for the equipment must be completely sure of its total safety and those involved. To carry out any type of maintenance, the main switch must be in the off position and locked with a padlock of up to 8mm. The equipment must also be safely depressurized.



10.1.2 Preventive maintenance routines

The table below shows an estimate for component replacement periods.

Preventive maintenance routines								
Equipment/Parts		Exchange Period Recommended by Raumak						
		Weekly	30d	60d	90d	12m	18m	24m
Vertical Seal Set	Adhesive Teflon	x						
	Teflon Backing		x					
	Silicone		x					
	Resistance		x					
	Headboard					x		
Horizontal Seal Set	Adhesive Teflon	x						
	Teflon Backing		x					
	Silicone		x					
	Resistance		x					
	Headboard					x		
	Rubber Profile			x				
	Shaft							x
	Sliding Bearings						x	
	Cutting Blade				x			
Set Track Film	Conveyor Belts				x			
Pneumatic	Coalescing Filter							x
	Filter Element							x
	Silencer							x
	Cylinder Repairs							x
Eletrical Panel	Cooler Filters	x						



10.1.3 Cleaning

Keeping the equipment clean is an important daily preventive maintenance task. This is a way to prevent product debris and particles from depositing on moving systems, causing premature wear or damaging the cycle, and also to prevent such debris from entering electrical or control systems causing damage or malfunction of the machine. When cleaning, pay attention to some machine conditions, such as clearances, wear of mobile systems, possible loose parts, pneumatic connections or possible air leaks, conditions of the electrical connection cables, i.e., take the opportunity to perform a general inspection of the equipment. If there are any anomalies, you can take the necessary measures to prevent breakages or major problems.

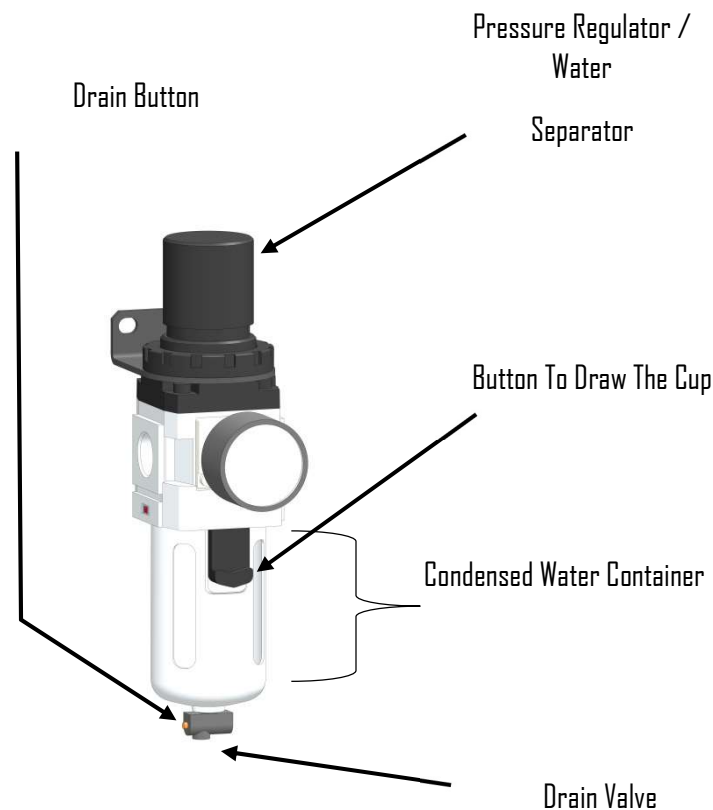


10.1.4 Draining water

The process of generating compressed air, through the condensation of gases contained in the air, tends to produce moisture and water, which in turn are very harmful to networks and pneumatic components. For this reason the machine is equipped with a filter, which will eliminate solid particles, oil contaminants and a part of the water. These will be retained in the cup and should be eliminated daily through the manual drain valve. Caution: never let the water level exceed the level indicated on the mixing bowl.

Drain by following the instructions below:

1. The drain valve has a button, when pressed, allows air, water and impurities contained in the cup to be expelled. Press it until all the water is expelled and only the air comes out of the valve. Then release it. This cleaning can be done with pressurized equipment;
2. Place a container under the drain valve to contain the waste;
3. Close the air inlet valve on the mains supply;
4. To discard the water from the cup, the serve by pressing the button and pulling down, rotate and remove the cup for cleaning. Then reassemble the cup;
5. Open the air line valve.





10.1.5 Pressure regulator filter

The filter of the pressure regulator must be inspected weekly evaluating its visual aspect (coloration) to verify if there is need of filtering element replacement. In case of saturation of this element, the replacement must be done. To make the change, press the button of fitting and pull the cup down unclipping it allowing the replacement. Afterwards perform the reverse procedure for assembly.

10.2 Lubrication

The parts of the machine that will need to be lubricated are:

- Gearmotors (According to tables extracted from the manufacturer's manual); however, the reducer model must be checked, since the L-line reducers used in most Raumak components do not require lubrication;
- Bearing bearings found in mats;
- Linear Guides;
- Table of approved lubricants for WEG CESTARI gearboxes:

Recommended Synthetic Lubricants		
Manufacturer	Viscosity mm ² /s (cSt) at 40°C - ISO -	Type of Lubricant Temp. maxim of Oil 90°
ExxonMobil	VG 460	SHC 634
Shell	VG 460	TIVELAS 460
Kluber	VG 460	KLUBER SYNTH – EG4-460
Kluber	VG 460	SYNTHESO D 460 EP
Petrobrás	VG 460	LUBRAX SYNTESIS GEAR A - 460
Castrol	VG 460	Syntethic A 460
Recommended Synthetic Lubricants		
L03 = 30ML	L04= 50ML	L05= 70ML

The oil change time is defined according to the operating temperature; see the table below:

Operating Temperature	OIL Mineral CLP	Synthetic Oil CLP HC hydrocarbons	Synthetic Oil CLP PG Polyglycol
80 °C	5000 horas	15000 horas	25000 horas
85 °C	3500 horas	10000 horas	18000 horas
90 °C	2500 horas	7500 horas	13000 horas
95 °C	2000 horas	6000 horas	8500 horas
100 °C	--- ---	3800 horas	6000 horas
105 °C	--- ---	2500 horas	4000 horas
110 °C	--- ---	2000 horas	3000 horas

10.2.1 Lubrication of rolling bearings

The bearing must be lubricated with mineral oil-based grease with quality K2K, DIN 51825. Apply grease using a grease gun.

Lubrication Interval

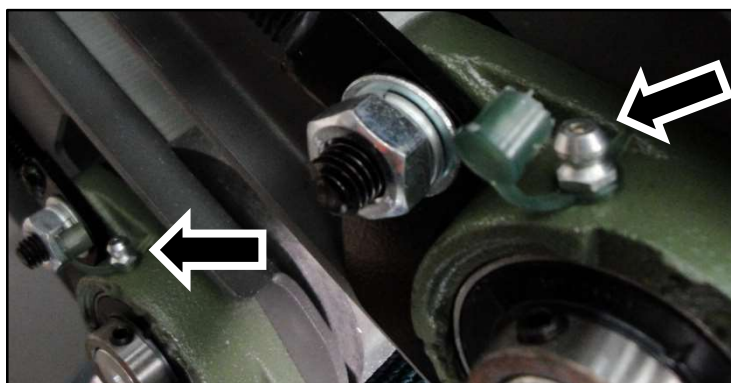
The lubrication interval varies according to the various factors and operating conditions of the equipment, such as: quality of the grease used, operating temperature in the bearing, rotation, applied load and environmental contamination (humidity, dust, salinity, gases, etc.). These factors, alone or in combination, affect the life of the grease and consequently the life of the rolling bearing. Table below:

Temp. oper. °C	Environmental conditions lubrication interval		
	Clean	Dirty	Dirty and Humid
50	3 years	6 months	3 months
70	1 year	2 months	1 month
100	3 months	2 weeks	1 week
120	6 weeks	1 week	3 days

Board 4.1.1.1

In many cases, verification of factors and operational conditions can be difficult. For this, we suggest the help of the lubricant supplier, who will be able to provide more detailed guidance on the application of the lubricant and the lubrication period.

For lubrication, a sufficient quantity of grease must be introduced through the grease pins, figure below, until a small portion of contaminated grease runs out of the bearing, between the seal and the inner ring or outer ring (depending on the type). sealing). It is extremely important that lubrication is done with the rolling bearing in operation, so that there is a uniform injection of grease throughout the entire bearing cavity.



Other bearings that are not shielded, perform a visual inspection and lubricate every 4 months as these have reduced wear compared to mats.

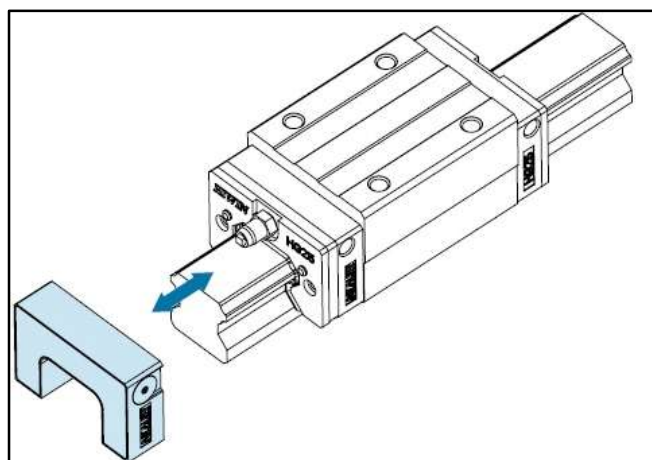
10.2.2 Lubrication of linear guides

In the traditional linear guide system, insufficient lubrication causes wear of the contact surfaces. With a linear motion guide, a centralized oil lubrication system is used, which channels the lubrication oil to the joint pipe.

Lubrication interval and type of lubricant

Each linear guide is lubricated with grease lithium soap base before shipment. In operation, we recommend that the guide is lubricated as indicated for each set interval measured in hours. You can make lubrication through the grease nipple. Generally, grease is applied for speeds not exceeding 60 m / min speeds exceeding the same, requires high viscosity oil as a lubricant.

The recommended oil viscosity is about **32~150cSt**. The oiler can be replaced by an automatic oil lubrication module as shown in figure 4.1a and 4.2b. Considering that oil evaporates faster than grease, the recommended oil feed rate is approximately 0.3 cm/hr.



4.1a



4.2



10.2.3 Moto reducers

Consistent in design, SEW Spiroplan® gearmotors are lubricated for life. If necessary, restore or renew the surface protection coating / anti-corrosion coating.

The following table from the manufacturer shows the time intervals to be adhered to and the corresponding measures

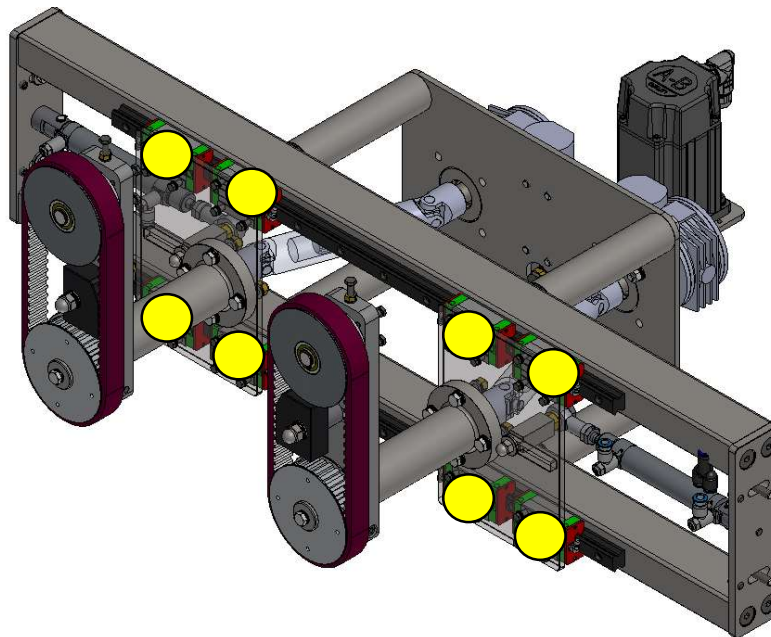
Frequency	To Do?
➤ Every 3000 operating hours, at least every 6 months	➤ Check oil and oil level ➤ Check for running noises indicating bearing damage. ➤ Visual inspection of seals (leaks). ➤ In gearboxes with torsion arm: check elastic bushing, replace when necessary
➤ □ Depending on operating conditions, at least every 3 years ➤ □ According to oil temperature.	➤ Change mineral oil
	➤ Replace the bearing grease (recommendation) ➤ Replace the seal (do not mount in the same position)
➤ Depending on operating conditions, at least every 5 years ➤ According to the temperature of the oil.	➤ Change synthetic oil
	➤ Replace the bearing grease (recommendation) ➤ Replace the seal (do not mount in the same position)
➤ Variable (depends on external influences)	➤ Touch up / redo the anticorrosive protection paint.

If necessary for more information on the maintenance of geared motors, it is recommended to consult the manufacturer's operating instructions manual (SEW).

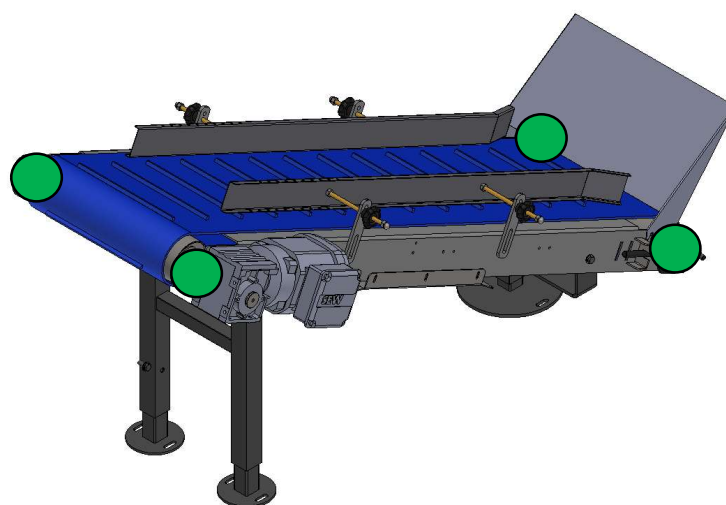


10.3 Lubrification Map

-  Linear guides
-  Bearings



Film Drive Belt



Out Put Conveyor



10.4 Spare Parts List

Spare Parts List - NS 4140		
Code	Description	Quantity
132557	SSR SCHNEIDER 530V 75A SSRPCDS75A2	1
1397	FUSÍVEL DE VIDRO PEQUENO 4A RETARDADO SLOW	5
1398	FUSÍVEL DE VIDRO PEQUENO 10A RETARDADO	5
2599	FUSÍVEL DE VIDRO 1A	2
250035	FACA PARA SOLDA RAIADA 370MM C/ 2 CIL. C/ EASY OPEN	1



10.5 Cause and Defect Ratio

In case of sudden defects, we guide you through the procedure that will enable you to find the defect. Read carefully each one of the topics and evaluate in which of them the problem fits. Below are listed the possible causes, allowing you to find and solve the problem.

10.5.1 Identify the symptom of the problem

- What's the matter with that?
- What were the circumstances when the problem occurred?
- Could climatic conditions have been a factor?
- What factors may have contributed?

10.5.2 Section the problem

- Look at the problem.
- In what area of the machine is the problem occurring?
- Has anything changed recently?

10.5.3 Isolate the problem

- First try simple things.
- Observe indications and problem codes.
- Check test points.

10.5.4 Initial procedure

- Check all fuses for a malfunction (indicated by the LED on).
- Check that all PLC and expansion outputs are on.

10.5.5 Machine does not turn on

- Check that the power cable is connected correctly.
- Check the circuit breaker.
- Check that the main switch is on.

10.5.6 Machine does not turn on

- Check that the power cord is connected correctly.
- Check the circuit breaker.
- Check that the master switch is on.



10.5.7 Jaw does not open or close

- Insufficient air pressure.
- Burnt valve coil.
- Damaged pneumatic valve.
- Flow controllers damaged.
- Damaged gag cylinder repair.
- Misaligned rulers.
- Time values do not agree.

10.5.8 Horizontal seal metal does not advance or recede

- Damaged pneumatic valve.
- Burnt valve coil.
- Damaged cylinder repair.
- Magnetic gag sensor not positioned correctly or damaged.
- Sensor triggered on jaw.

10.5.9 Plastic stretches and does not cut

- Lack of pressure in the jaw.
- Lack of support in the jaw (rubbers).
- Overheating.
- Knife damaged or without cutting edge.
- Defect in packaging.

10.5.10 Inefficient Seal

- Insufficient air pressure.
- Support metal out of adjustment.
- Broken heating element.
- Defective temperature controller.
- Broken or bad contact wire rope or terminations.
- Damaged solid-state relay.
- Check welds power percentage in the IHM parameters.

10.5.11 Do not seal top or bottom of package

- Insufficient air pressure.
- Support metal out of adjustment.
- Broken resistance.
- Lack of gag support.
- Misaligned metals.
- Check welds power percentage in the IHM parameters.



10.5.12 Resistor do not heat up

- Broken solder fuse.
- Problem transformer.
- Bad contact in the electric cables.
- Broken heater.
- Function 'welds' disabled.
- Check welds power percentage in HMI parameters.

10.5.13 Package breaks easily when soldering

- Inefficient cooling.
- Problem in the package.
- Horizontal weld too hot.
- Disregulated backing metal.

10.5.14 Variation in package length

- Packing too loose or jammed.
- Unadjusted length.
- Photocell too close or too far away.
- Defects or printing on the line of the 'stripes' triggering the sensor improperly.

10.5.15 When the machine is running, the air pressure drops

- Air compressor does not provide sufficient pressure.
- Leakage in the compressed air line.
- Clogged filter elements.
- Partially closed or fully closed damper valves.

In case of doubt, we have the department of "ASTEC" (technical assistance by telephone), which is composed of specialized technicians who will provide the necessary support for the customer to perform the desired service.

We would like to emphasize that Raumak gives courses and training at the factory in Jaraguá do Sul - SC.

For more information, please contact us by phone 55 47 3370 4540. Extension 202 or via e-mail **service@raumak.com.br** - **To the care of the Technical Assistance.**